NumPy Practice Questions

Topic: 1D and 2D NumPy Arrays

1. Write a program to create a 1D NumPy array with five elements entered by the user.

```
import numpy as np
arr = []
for i in range(1,6):
  arr.append(int(input(F"Enter the {i} element : ")))
arr1 = np.array(arr)
print(arr1)
```

```
Enter the 1 element : 1
Enter the 2 element : 2
Enter the 3 element : 3
Enter the 4 element : 4
Enter the 5 element : 5
[1 2 3 4 5]
```

2. Write a program to find the length of a 1D NumPy array.

```
import numpy as np
arr = np.array([1,2,3,4])
print("Length of the arr : ", len(arr))
```

```
Length of the arr :  4
```

3. Write a program to access the first, middle, and last element of a NumPy array.

```
import numpy as np
arr = np.array([1,2,3,4,5,6,7])
print(f"First element : {arr[0]}")
print(f"Middle element : {arr[(len(arr)//2)]}")
print(f"Last element : {arr[-1]}")
```

```
First element : 1
Middle element : 4
Last element : 7
```

4. Write a program to slice a NumPy array from index 2 to 5.

```
import numpy as np
arr = np.array([1,2,3,4,5,6,7,8])
print(arr[2:6])
```

```
[3 4 5 6]
```

5. Write a program to print all elements of a NumPy array using a loop.

```
import numpy as np
arr = np.array([1,2,3,4,5,6,7])
for i in arr:
  print(i)
```

```
1
2
3
4
5
6
7
```

6. Write a program to add 10 to each element of a NumPy array.

```
import numpy as np
arr = np.array([1,2,3,4,5])
print(arr+10)
```

```
[11 12 13 14 15]
```

7. Write a program to find the sum and average of elements in a 1D NumPy array

```
import numpy as snp
arr = snp.array([1,2,3,4,5])
print(f"Sum all elements of arr : {np.sum(arr)}")
print(f"Average of all elements of arr : {np.average(arr)}")
```

```
Sum all elements of arr : 15
Average of all elements of arr : 3.0
```

8. Write a program to find the maximum and minimum value in a NumPy array.

```
import numpy as np
arr = np.array([1,2,3,4,5,6,7])
print(f"Maximum value in arr : {np.max(arr)}")
print(f"Minimum value in arr : {np.min(arr)}")
```

```
Maximum value in arr : 7
Minimum value in arr : 1
```

9. Write a program to reverse a NumPy array using slicing.

```
import numpy as np
arr = np.array([1,2,3,4,5,6])
print(arr[::-1])
```

```
[6 5 4 3 2 1]
```

10. Write a program to check whether a specific element exists in a NumPy array

```
import numpy as np
arr = np.array([1,2,3,4,5,6])
ele = int(input("Enter the element to check : "))
if ele in arr:
  print("Element exists in arr!")
else:
  print("Element does not exist in arr!")
```

```
Enter the element to check : 66
Element does not exist in arr!
```

B. 2D NumPy Array Questions

1. Write a program to create a 2D NumPy array of size 3×3 and display it.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

2. Write a program to find the shape of a 2D NumPy array.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr.shape)
```

```
(3, 3)
```

3. Write a program to access a specific element from a 2D array using row and column index.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Element at row 2 column 1 is : ", arr[2,1])
```

```
Element at row 2 column 1 is :  8
```

4. Write a program to extract the first row and first column from a 2D NumPy array.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
```

```
print("First Row : ", arr[0])
print("First Column : ", arr[:,0])
```

```
First Row :  [1 2 3]
First Column :  [1 4 7]
```

5. Write a program to slice a 2D NumPy array to get a sub-matrix.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Submatrix : \n", arr[1:,1:])
```

```
Submatrix :
 [[5 6]
 [8 9]]
```

6. Write a program to calculate the sum of all elements in a 2D NumPy array.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Sum of all elements in arr : ", np.sum(arr))
```

```
Sum of all elements in arr :  45
```

7. Write a program to calculate row-wise sum of a 2D NumPy array.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Sum of row-wise elements in arr : ", np.sum(arr, axis = 1))
```

```
Sum of row-wise elements in arr :  [ 6 15 24]
```

8. Write a program to calculate column-wise average of a 2D NumPy array.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Sum of column-wise elements in arr : ", np.sum(arr, axis = 0))
```

```
Sum of column-wise elements in arr :  [12 15 18]
```

9. Write a program to multiply each element of a 2D NumPy array by 2.

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr*2)
```

```
[[ 2  4  6]
 [ 8 10 12]
 [14 16 18]]
```

10. Write a program to find the maximum value from each row of a 2D NumPy array

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Maximum value from each row in arr : ", np.max(arr, axis = 1))
```

```
Maximum value from each row in arr :  [3 6 9]
```